

			$0.100 \times 10^{-3} \text{ kg}$	A1
4	(a)	(i)	$v = f\lambda$ $0.90 = f(0.30)$ $f = 3.0 \text{ Hz}$	M1
		(ii)	$f = \frac{1}{2\pi} \sqrt{\frac{28}{m}}$ $m = 0.0788 \text{ kg}$	A1
		(iii)	1. <u>Driving force is larger</u> due to the larger amplitude of wave, OR more energy transfer.	M1

			Amplitude of the vertical oscillations will also increase.	A1
		2.	Wavelength of wave increases. Since the speed remains the same, by $v = f\lambda$, driving frequency of wave will decrease. Since driving frequency is not equal to natural frequency, <u>amplitude will decrease</u> (no more resonance)	M1 A1
		3.	Damping force increases. Therefore, amplitude decreases.	M1 A1
	(b)	(i)	Light in which the oscillations of the electromagnetic fields <u>are all in a single plane.</u>	A1
		(ii)	The displacements due to the two waves are in perpendicular axes, thus their <u>vector sum will not be able to produce</u> distinct maxima and <u>minima</u>, since they can <u>never be</u> in the same or <u>in opposite directions</u> (OR there will not be complete cancellation). Hence, the contrast between bright and dark fringes is not observable.	M1 A1
		(iii)	For X, after the initial unpolarised light passed through the polariser, the light became plane polarised with $I_x = \frac{1}{2}I$ with amplitude $A_x = A$ (i.e. unchanged for any θ value of X). Hence, $I_z = I_x (\cos^2 45^\circ)(\cos^2 45^\circ)$ $= \frac{I}{2} \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)$ $= \frac{I}{8}$	M1 M1 A1
5	(c)	(i)	Correct directions with line of action of force vector passing	A1